Association for
Project Management

APM Project Management Qualification

Learner Study Pack

Because when projects
succeed, society benefits

Requirements Management

Introduction

This section of the Learner Study Pack will cover the Assurance competence of the APM Project Management Qualification.

This section will enable you to gain an understanding of Requirements Management as the process of capturing, assessing and justifying stakeholders' wants and needs.

Learning objective and outcomes

Learning objective:

16. Understand requirements management as the ability to capture and monitor the requirements of a project.

By the end of this activity, you should be able to:

16a) Understand how to establish scope through requirements management processes (such as gather, analysis, justifying requirements and baseline needs).

16b) Understand how to manage scope through configuration management processes (such as planning, identification, control, status accounting and verification audit).

Study time

This section of the Learner Study Pack will take approximately 1 - 2 hours to complete.

Recommended resources

In addition to your Learner Study Pack, use the below references for more information on Assurance:

- APM Body of Knowledge Seventh edition 4.1.2 (Objectives and requirements)
- APM (2015) Requirements Management by Matthew Bell https://www.apm.org.uk/news/requirements-management-by-matthew-bell/
- APM 2020, Understanding requirements, the basis of delivering quality Accessed at: https://www.apm.org.uk/blog/understanding-requirements-the-basis-of-delivering-quality/

Overview

Requirements management is an ongoing process throughout the project life cycle and should be commenced in the initial phase of the project. It facilitates the development of an agreed scope management.

It is important to go through these processes:

- **Gather**

  After having identified key stakeholders as far as possible, information needs to be gathered from these to ensure that all possible requirements are gathered, recorded and documented.

  Where some stakeholders are missed from these early discussions, this can lead to important information being missed or to conflict further down the line.

- **Analysis**

  Function analysis or function cost analysis will provide detail on the value that the requirements will contribute to the organisation.

  Analysis should enable any gaps to be identified or conflicting requirements resolved. This can also provide information on alternative solutions.

  The outcomes of the analysis should be fed back to stakeholders and consensus reached.

- **Justifying Requirements**

  Functional requirements should be prioritised, which link directly back to the benefits, using techniques such as MoSCoW (M – Must have; S – Should have; C – Could have; W – Won't have).

  Reporting on the outcomes to stakeholders is then required to mitigate any misunderstandings or disputes.

- **Baseline Needs**

  Once the functional requirements have been set, these form the baseline requirements for the project from which the project manager and project team can determine the project deliverables, manage change and mitigate scope creep.

Function Analysis

We should only be willing to accept benefits at a cost that makes them worthwhile to an organisation thus creating value.

Value however must be created at the beginning of the project through the development of the business case and in the selection of the option to be delivered, and then be maintained throughout the project using, for example, cost and schedule management.

Function analysis or function cost analysis are value-based techniques which allow the analysis of the requirements and their business drivers by combining information from functions such as schedule management and investment appraisal.

Analysing the time and cost required to deliver the requirements will result in thorough understanding of requirements and the value they will contribute to the overall objective.

This step of the process also tests any assumptions made as part of the analysis primarily providing feedback to stakeholders, building consensus and generating ideas.

The results of the analysis are communicated through individual consultation or group workshops. This leads to a debate about functionality and alternative ideas.

The product owner has a vital role in ensuring that requirements are always seen relative to business needs and advises on the acceptance criteria necessary.

The result is a baselined set of options for functional requirements. These can then be used to examine the value of alternative solutions during solutions development.
Value represents the amount of benefit that will accrue from the requirement and could be used to justify its priority – how important this requirement is compared to the others.

Prioritisation Techniques

Sometimes, more requirements are requested than it is feasible to deliver, so a prioritisation exercise is needed to highlight the most essential requirements and justify why the chosen requirements should be the ones that are baselined.

A common prioritisation technique is the MoSCoW approach:

(M) Must have

(S) Should have

(C) Could have

(W) Won't have

If using an agile methodology, the 'could have' and 'should have' requirements would be sacrificed if at any time the project was predicted to go over budget or be late. The 'must haves' would only be reduced as a very last resort.

The main benefit of the process is to ensure that essential requirements are understood as an input to the work to select the optimal solution and then define the detailed scope of work to be delivered with acceptance criteria.

This level of rigour mitigates the risk that in later life cycle there is dispute between the project and the organisation about completion and transition of the deliverables into use.

Configuration Management Processes

Scope management is closely linked to configuration management by planning, protecting and controlling changes to requirements.

The activities of configuration management contribute to scope management as follows:

- **Planning**

  This stage defines process and procedures along with setting clear roles and responsibilities to communicate requirements that team members need to follow.

- **Identification**

  This stage divides the project into configurable items each with a unique reference so that the project manager and team and identify the deliverables of the project.

- **Control**

  This stage seeks to ensure that changes to these items are controlled and interrelationships between the items are understood to fully understand the impact of the change.

- **Status Accounting**

  This ensures traceability of configuration items throughout their development.

  This supports record keeping and document control for the project thus maintaining a robust audit trail of configuration records over the course of the project.

- **Verification Audit**

  This ensures that the deliverable conforms to its requirements and configuration information and that completed deliverables match those documented.

  Audits should be undertaken throughout the life cycle to maintain full end-to-end configuration management of the project.

Scenario One:

The ‘Snappy Appy’ renovation project involves multiple stakeholders with very diverse requirements and expectations.

As the Project Manager you may find that as the project progresses, you encounter challenges related to the requirements management of the overall project.

What steps could you follow to identify the requirements at the start of the project?

What techniques could you use to establish the order of priority for the varying requirements?

How can effective requirements management contribute to the success of the project and the satisfaction of the stakeholders involved?

Why is it important for you to establish the baseline scope of the project?

Scenario Two:

You know better than anyone how closely scope management and configuration management are linked and so you've decided to consider examples of the kinds of things that could be involved at each of the 5 steps of configuration management.

What are some examples of the different considerations that could be involved in the Planning, Identification, Control, Status Accounting or Verification Audit phases of the 'Snappy Appy' renovation project?

Learning review questions

Before you attempt the exam practice questions in the next section, we recommend you review your learning to quickly see which aspects you don't understand or can't remember as well.

It will also boost your confidence by confirming what you have learnt and your demonstrated ability to apply that knowledge, so that you are ready to practise for the exam.

Here are some questions you may wish to answer to review your learning from the Requirements Management section:

- What is an example of a technique that may enable a project manager to define the requirements for a project?
- Why is stakeholder identification so important to defining the scope for a project?
- How can reporting back on scope development to stakeholders help to mitigate conflict or misunderstandings during a project?
- Why is there a strong link between scope management and configuration management?

Exam practice questions

16.1 You are managing a project to develop a new software solution for a financial services company. The Project Sponsor has provided you with a long list of requirements, but you both agree that you cannot deliver all of them within the project's timeline. Therefore, they need to be prioritised.

Which of these is the best factor to use when determining the prioritisation of requirements?
(1 mark)

a. The opinion of key stakeholders.
b. The complexity of the solution.
c. The benefit/value to the end user.
d. The resources required to develop the solution.

16.2 You are the project manager for a software development project, and the client has proposed a change to the requirements. The change involves adding a new feature that wasn't originally planned. The client believes this addition will significantly enhance the project deliverable.

As the project manager, how should you respond to this proposed change?
(1 mark)

a. Immediately pause the project until the new requirements have been analysed.
b. Accept the new requirements without further analysis as the client is funding the project.
c. Assess the impact of the new requirements, considering factors such as cost, quality, and time.
d. Accept the new requirements but encourage the client to not propose any more as this will create further delay.

16.3 Your project sponsor has emailed to ask you to explain a configuration management process, so that they can understand what's involved at each step.

Explain the five steps in a configuration management process.
(5 marks)

16.4 You are the project manager for a commercial development project and you have now gathered all requirements.

In the analysis stage, you decide to perform a Function Cost Analysis in order to identify the **_____(a)_____** of the requirements to the organisation.

You then use the **____(b)______** technique to priortise the requirements.

(2 marks)

(a)
1. Cost
2. Value
3. Investment
4. Risk

(b)
1. SMART
2. SWOT
3. PESTLE
4. MoSCoW

Exam practice answers

Question	Answer / mark scheme
16.1	C
16.2	C
16.3	1 mark for a correct explanation of each of the below steps, 5 marks available in total. No marks will be given for only stating the name of each step. • Planning: Involves defining the procedures / processes to follow, with roles and responsibilities. • Identification: Involves breaking down the project into items and giving each a unique reference. • Control: Ensures that any changes to items are controlled and dependencies between items are understood. • Status Accounting: Items are under full version control so there is an audit trail of all configurable items. • Verification Audit: Ensures that all items conform to requirements / that deliverables meet what has been documented.
16.4	(a) 2 (b) 4